

Classwork 06//KL Divergence

Student Name: _____ **Points:** 30

A Number: _____ **Due Date:** March 19, 2026

Consider a random variable $X \sim \text{Lognormal}(\mu, \sigma^2)$ with the probability density function (pdf):

$$p(x) = \frac{1}{x\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln x - \mu)^2}{2\sigma^2}\right), \quad x > 0$$

and normal distribution $Y \sim \mathcal{N}(m, v^2)$ with pdf

$$q(x) = \frac{1}{v\sqrt{2\pi}} \exp\left(-\frac{(x - m)^2}{2v^2}\right), \quad x \in \mathbb{R}$$

The KL divergence formula is

$$KL(p||q) = \int_0^\infty p(x) \log \frac{p(x)}{q(x)} dx$$

which is defined only if the support of P is contained in the support of Q . This is very important. Noting the above, compute the KL divergence $KL(\text{Lognormal}(\mu, \sigma^2) || \mathcal{N}(m, v^2))$.